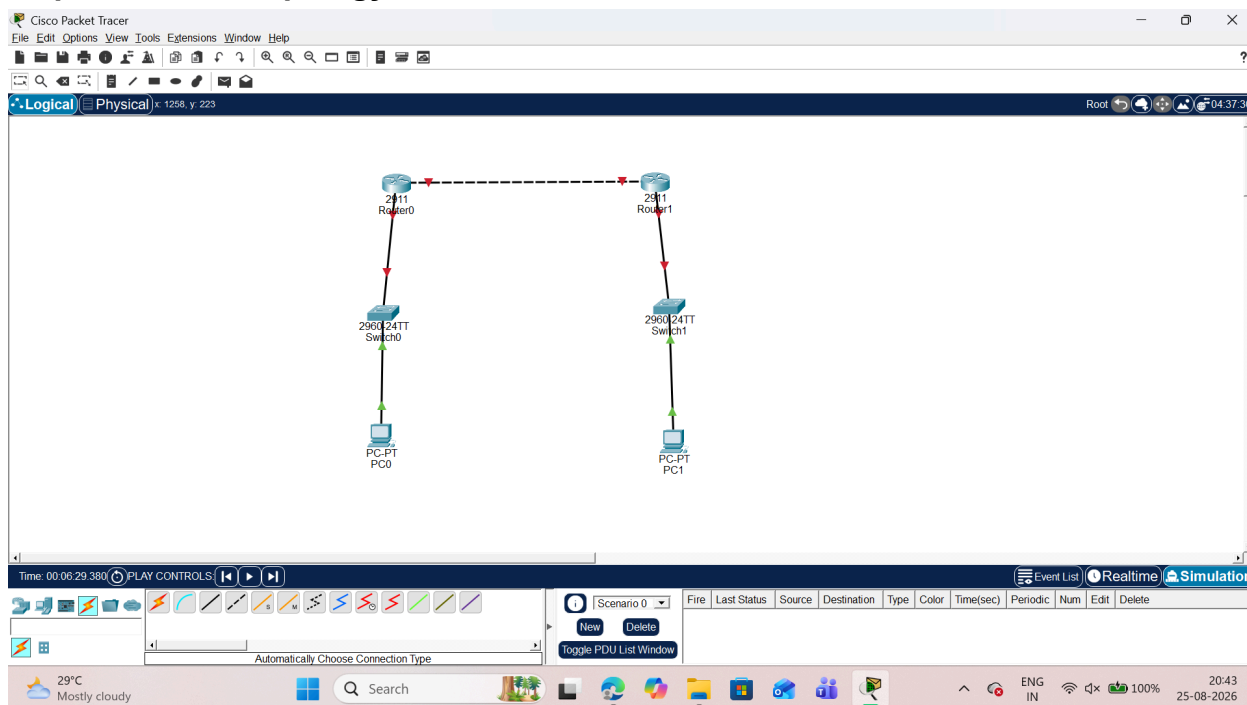


PRACTICAL - 6

AIM:IP Security (IPsec) Configuration

Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

Step 1. Network topology



Step 2. Configure PC IP addresses a.PC0

The screenshot shows the configuration window for PC0. The 'IP Configuration' tab is active, displaying the following settings for the FastEthernet0 interface:

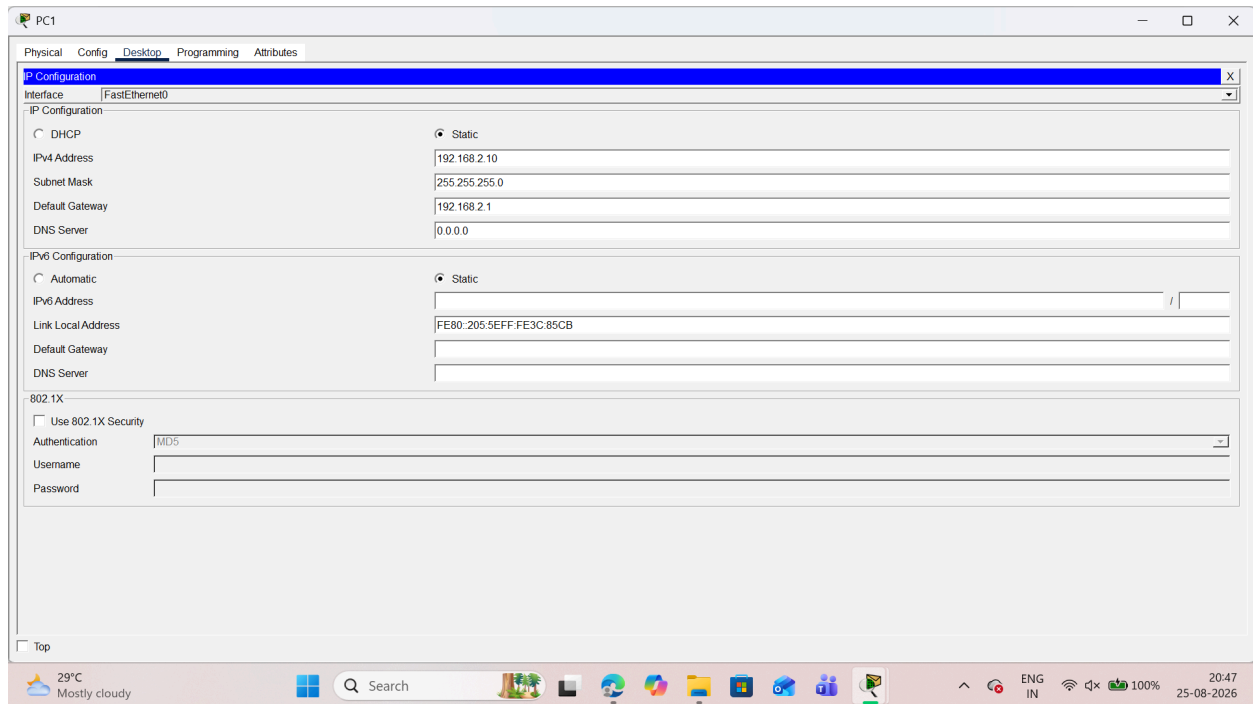
- IP Configuration:**
 - ☐ DHCP
 - ☒ Static
 - IPv4 Address: 192.168.1.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1
 - DNS Server: 0.0.0.0
- IPv6 Configuration:**
 - ☐ Automatic
 - ☒ Static
 - IPv6 Address: (empty)
 - Link Local Address: FE80::201:42FF:FE39:CB3D
 - Default Gateway: (empty)
 - DNS Server: (empty)
- 802.1X:**
 - ☐ Use 802.1X Security
 - Authentication: MD5
 - Username: (empty)
 - Password: (empty)

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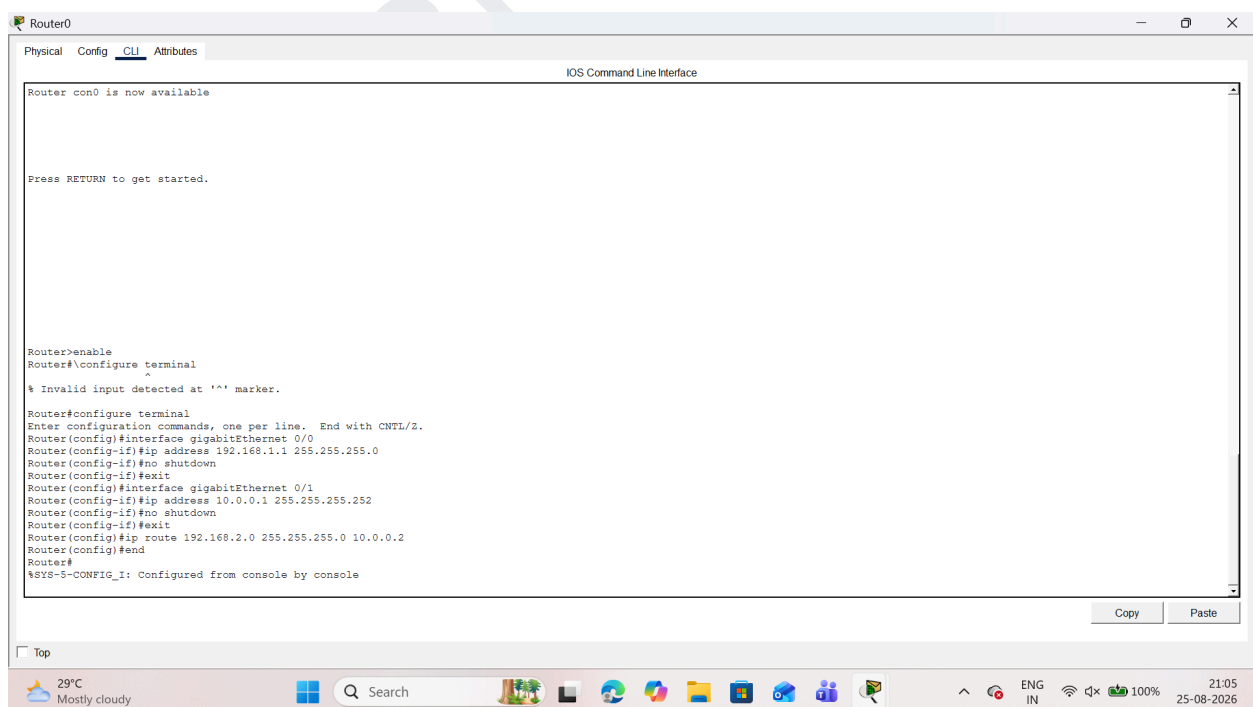
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b.PC1



Step 3. Configuring the Routers

a.Router 0

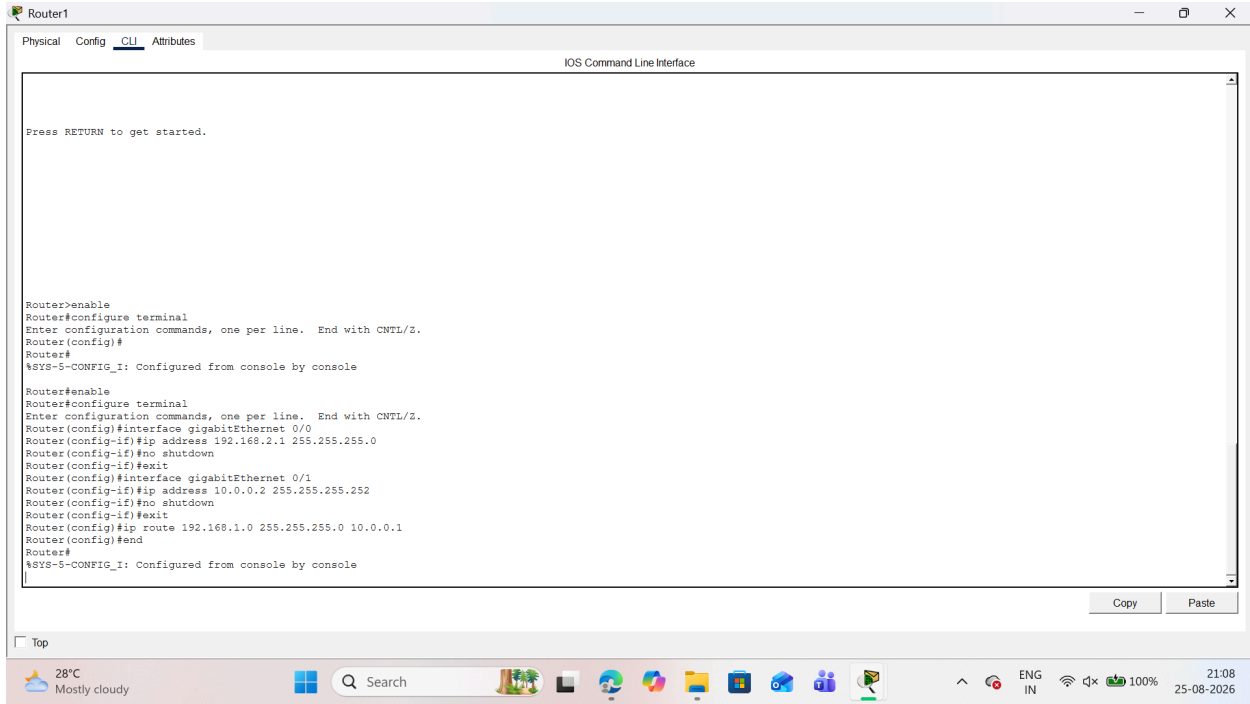


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b.Router 1



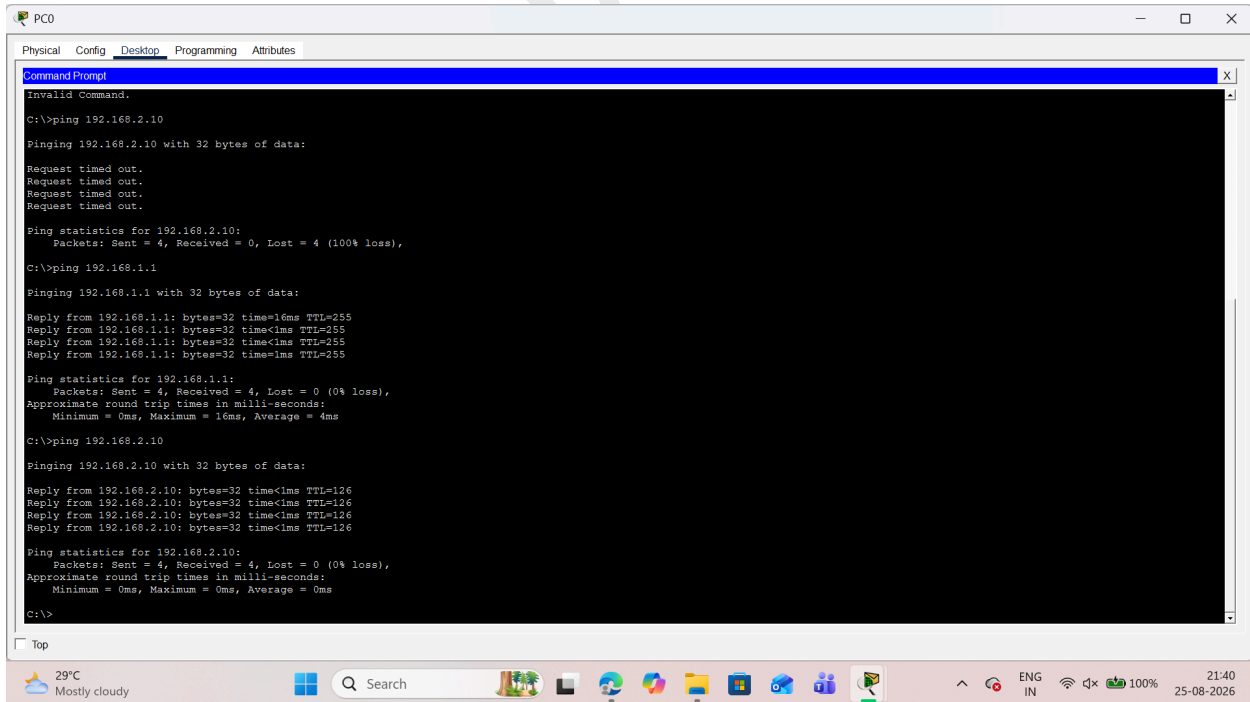
```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 4. First test connectivity using ping command



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Invalid Command.

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=16ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 4ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

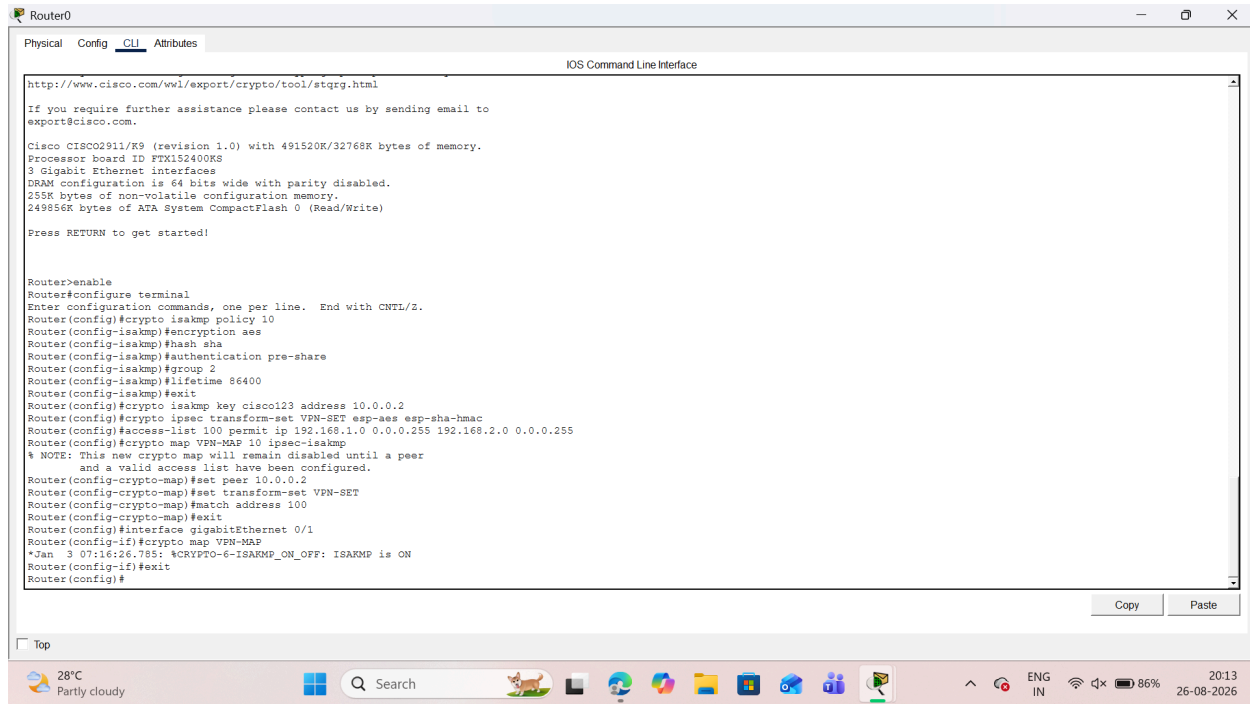
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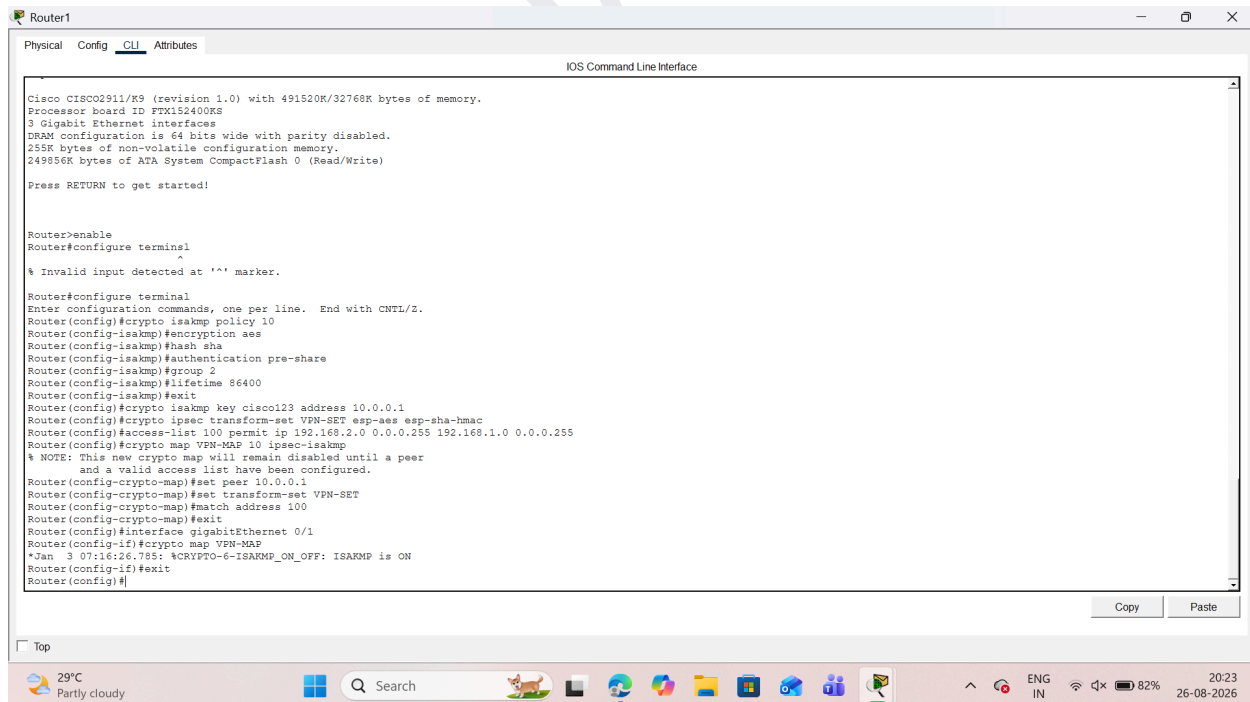
Step 5 . Configure IPsec on Router 0



The screenshot shows the CLI interface of Router0. The configuration commands entered are as follows:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.2
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.2
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#
```

Step 6 Configure IPsec on Router 1



The screenshot shows the CLI interface of Router1. The configuration commands entered are as follows:

```
Router1>enable
Router1#configure terminal
% Invalid input detected at '^' marker.
Router1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#crypto isakmp policy 10
Router1(config-isakmp)#encryption aes
Router1(config-isakmp)#hash sha
Router1(config-isakmp)#authentication pre-share
Router1(config-isakmp)#group 2
Router1(config-isakmp)#lifetime 86400
Router1(config-isakmp)#exit
Router1(config)#crypto isakmp key cisco123 address 10.0.0.1
Router1(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router1(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router1(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router1(config-crypto-map)#set peer 10.0.0.1
Router1(config-crypto-map)#set transform-set VPN-SET
Router1(config-crypto-map)#match address 100
Router1(config-crypto-map)#exit
Router1(config)#interface gigabitEthernet 0/1
Router1(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router1(config-if)#exit
Router1(config)#
```

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Step 7 Test the IPsec tunnel and verifying

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router con0 is now available

Press RETURN to get started.

Router>show crypto isakmp sa
^
% Invalid input detected at '^' marker.

Router>enable
Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst      src      state      conn-id slot status
10.0.0.2  10.0.0.1  QM_IDLE    1028     0  ACTIVE

IPv6 Crypto ISAKMP SA

Router#
```

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

GigabitEthernet0/1  10.0.0.1  YES NVRAM  up      up
GigabitEthernet0/2  unassigned YES NVRAM  administratively down down
Vlan1               unassigned YES NVRAM  administratively down down

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
L       192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
L       192.168.2.0/24 [1/0] via 10.0.0.2

Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst      src      state      conn-id slot status
10.0.0.2  10.0.0.1  QM_IDLE    1028     0  ACTIVE

IPv6 Crypto ISAKMP SA

Router#show crypto ipsec sa
interface: GigabitEthernet0/1
Crypto map tag: VPN-MAP, local addr 10.0.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 6, #pkts encrypt: 6, #pkts digest: 0
```

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```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#show crypto ipsec sa
interface: GigabitEthernet0/1
Crypto map tag: VPN-MAP, local addr 10.0.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags=(origin_is_acl,)
#pkts encaps: 6, #pkts encrypt: 6, #pkts digest: 0
#pkts decaps: 5, #pkts decrypt: 5, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 10.0.0.1, remote crypto endpt.: 10.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0xB5BC0813(3048998931)

inbound esp sas:
spi: 0xE9B03E6C(3920641644)

Router#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
Peer = 10.0.0.2
Extended IP access list 100
access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Current peer: 10.0.0.2
Security association lifetime: 4608000 kilobytes/3600 seconds
PFS (Z/N): N
Transform sets=
VPN-SET,
Interfaces using crypto map VPN-MAP:
GigabitEthernet0/1

Router#
```

Top

29°C Partly cloudy Search 20:37 26-08-2025

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